When the Gradient Sees Rank: Provable Necessity, Causal Recruitment, and Exact Composition in Trained Matrix Memories

Author Name(s) TBD†
Affiliation TBD
email@TBD

Abstract

A companion negative result showed that a matrix-valued latent bolted onto a vector-pretrained continuous-reasoning model never uses the rank of its state: truncating the latent to rank k leaves ProsQA accuracy flat across four training regimes and four readouts. That result left open whether the gradient is rank-indifferent in general or whether the tested task simply admitted a rank-1 solution. We answer with three from-scratch, matrix-native experiments whose readout is pinned to *exact continuous recovery* – never argmax – so that the rank-$m \to md$ argmax-capacity construction of Nichani et al. [16] cannot silently satisfy a rank bound the model never earned. (1) On an associative-memory task with a provable $\mathrm{rank}(Z) \geq K$ necessity and a single-matrix-state bottleneck that forecloses the position-decomposition escape (verified by a gradient-based blank-out test, not a shape check), gradient descent trained from scratch develops effective rank $\approx K$ (Spearman $\rho = 1.0$ at $d = 16$) and a causal force-rank step at $k \approx K$ (razor-sharp at $d = 8, K = 4$: rank $\leq 3 \Rightarrow 0.0$ recovery, rank $4 \Rightarrow 0.97$). (2) On a compositional extension requiring the same operator to compose under repeated self-application Z^h, 4/5 seeds recover exactly (cosine 1.00) to depth $h = 21$. An entity-subspace decomposition resolves an apparent contradiction – whole-matrix rank inflates to $\approx d$ – by showing the operator *restricted to the K-dimensional key subspace* is exactly a scaled cyclic permutation with restricted rank exactly K, while a full-rank, dynamically invisible complement absorbs every unconstrained degree of freedom. A deliberately rank-starved ($k = K - 1$) operator predicts its own depth-decay curve from its spectrum alone to within 0.02 at every hop – iteration depth is a sharper spectral-exactness probe than any single-hop tolerance. (3) Scaling d at fixed encoder width reveals a genuine *exactness* frontier, not the trainability wall a first pass suggested: every tested seed eventually transitions (late, seed-stochastic), but converged recovery degrades monotonically with d (cosine ≈ 1.0 at $d = 16$; 0.72 at $d = 48, K = d/4$; 0.39 at $d = 64, K = d/4$). Three independent axes of this program each declared a cell "dead" at a fixed step budget before a 2–2.5× extension reversed the verdict – a methodological finding we name, quantify, and caveat (a genuine budget-independent plateau also occurs, and does not always resolve in a result's favor). All experiments are synthetic and ≤1M parameters; we report this scope plainly and state what would falsify each claim. A concurrent, preliminary result on a production fast-weight architecture (DeltaNet) shows perfect unconstrained-arm composition and exact, uninflated rank recruitment at 2.5× tier budget, with the causal force-rank staircase still in flight.

1 Introduction

Continuous chain-of-thought (CoT) models feed a latent vector back into the residual stream instead of decoding an explicit reasoning token [8]. Matrix-valued variants of this idea introduce a structural observable that a vector latent does not have: the *rank* of the latent matrix Z. If Z carries multiple reasoning paths in superposition, rank should track how many paths are held, and truncating Z to rank k should hurt accuracy once k falls below what the task needs.

Our companion paper, *The Gradient Does Not See Rank* [10], tested this directly on a matrix-CODI model (a $d \times d$ bottleneck bolted onto a vector-pretrained GPT-2, trained by CODI-style distillation on ProsQA) and found the probe uninformative: the rank-k truncation curve is flat to within 0.6 percentage points across four training regimes and four readouts, including three explicitly nonlinear-in-Z readouts designed to rule out a constant-Jacobian artifact. Three training seeds reached accuracies of 81.5 ± 1.2pp while their final effective ranks spanned $\{4, 12, 13\}$ – the loss rewarded no particular rank. That paper closed the readout-mechanism question but left one confound explicitly open (its §7): ProsQA has a unique positive answer and a single distractor, so the flat curves are also consistent with the task being *rank-1-solvable* – in which case rank-blindness says nothing about the gradient's ability to use rank when a task actually requires it.

This paper closes that confound and asks the question the negative result could not answer on its own:

> *When a task's exact solution provably requires $\mathrm{rank}(Z) \geq K$, does gradient descent, trained from scratch under a hard matrix bottleneck, (a) develop effective rank $\approx K$, and (b) make that rank causally necessary?*

Answering this honestly requires removing every shortcut a model could use to appear rank-aware without actually needing rank: an argmax readout that tolerates interference (§2), a decoder that can read raw inputs instead of only the bottleneck (§2), and a fixed step budget that mistakes a slow transition for a wall (§5). We built three experiments, each pre-registered with decision criteria before any GPU ran, that close these shortcuts in turn.

Contributions.

1. **A provable** $\mathrm{rank}(Z) \geq K$ **lower bound with a readout discipline that keeps it valid.** Exact recovery of K independent key→value bindings from a linear memory needs rank K – a classical fact [1, 9] – but it holds only under *exact continuous* recovery. Under discrete argmax decoding, a rank-m matrix recovers $\approx md$ associations [16], and the bound silently collapses. §2 pins the readout to the literal unbind $\mathrm{pred} = Z \cdot \mathrm{key}_j$ scored by an absolute cosine bar, and closes the position-decomposition escape (any full-attention model can hold K items at K rank-1 positions) with a single-state P=1 bottleneck verified by a gradient-based blank-out test.

2. **Rank tracks** K **and is causally necessary** (§3). At $d = 16$, learned effective rank tracks K almost exactly (Spearman $\rho = 1.0$), and train-time force-rank training shows a step at $k \approx K$: at $d = 8, K = 4$, force-rank ≤ 3 gives 0.0 recovery and force-rank 4 gives 0.97.

3. **Exact composition to depth** $h = 21$**, and the invariant-subspace mechanism that explains it** (§4). A rank-sufficient trained operator composes exactly under 21-fold self-application; an entity-subspace decomposition shows the operator restricted to the K-dimensional key subspace is exactly a scaled K-cycle with restricted rank exactly K, and the apparent whole-matrix rank-inflation to $\approx d$ is a dynamically inert complement no real query ever visits.

4. **Depth as a spectral-exactness amplifier.** A rank-$(K{-}1)$ operator looks correct at shallow hops (cosine ≈ 0.94, comfortably above the 0.9 recovery bar) and is exposed only by 21-fold composition (recovery collapses to 0.06) – a sharper rank test than any single-hop tolerance, confirmed by predicting the entire depth-decay curve from the operator's own eigenspectrum, no raw keys needed.

5. **The exactness frontier, and a methodological finding about declaring cells dead** (§5). Scaling d at fixed encoder width does not hit a trainability wall – every tested seed eventually transitions – but converged exactness degrades monotonically with d. Three independent axes of this program each called a cell "dead" at a fixed budget before a 2–2.5× extension reversed the verdict; we name this the *three-budget-artifacts pattern* and give the counter-example (a genuine, budget-independent plateau) that keeps the rule from being a license to always re-test.

Scope, stated up front. Every experiment here is synthetic and ≤1M parameters – deliberately small, chosen so that the ground-truth minimal solution is provable rather than assumed. §7 argues why that scope is still informative and what it does not show. We do not claim these results demonstrate that rank helps *real* reasoning; that transfer step is explicitly future work. A brief, clearly-marked preliminary result on a production fast-weight architecture (DeltaNet) is reported only to bound what is already visible in a concurrent, unfinished campaign (§7); it is not this paper's headline result.

2 Setup: A Provable Bound and the Readout Discipline That Keeps It Valid

2.1 The associative-memory task

Each sample presents K key→value bindings, then queries one:

$$\mathrm{BIND}\ k_1 v_1 \ \cdots\ \mathrm{BIND}\ k_K v_K\ \ \mathrm{QUERY}\ k_j \ \rightarrow\ v_j.$$

Keys and values are fresh per sample – i.i.d. Gaussian, L_2-normalized (near-orthogonal), or exactly orthonormal via Haar-corrected QR sign correction [13] – so no static-vocabulary shortcut exists and generalization can be evaluated on never-seen bindings. Bind order is shuffled; the queried index is uniform over the K bindings. We call this *Task D*.

2.2 The provable lower bound

Require exact recovery $Z \cdot k_j = v_j$ for all $j = 1, \ldots, K$, with $\{k_j\}$ and $\{v_j\}$ each linearly independent (generic for random vectors, $K \leq d$). Stacking keys as columns of $K_{\mathrm{mat}} \in \mathbb{R}^{d \times K}$ and values as V_{mat}, exact recovery requires $Z \cdot K_{\mathrm{mat}} = V_{\mathrm{mat}}$, so

$$\mathrm{rank}(V_{\mathrm{mat}}) = \mathrm{rank}(Z \cdot K_{\mathrm{mat}}) \leq \mathrm{rank}(Z),$$

and since $\mathrm{rank}(V_{\mathrm{mat}}) = K$,

Fact 1 (Rank necessity)**.** $\mathrm{rank}(Z) \geq K$.

This rules out the rank-1 "vector-in-a-costume" shortcut $Z = uv_0^\top$ by contradiction for any $K > 1$: a rank-1 matrix cannot satisfy $K > 1$ independent exact constraints. The fact that exact recovery of K orthonormal key→value pairs from a linear memory needs rank K is classical [1, 9]; we do not claim it as novel. What is not classical, and what §3–4 measure, is whether a *trained, bottlenecked, matrix-native transformer spontaneously discovers* this rank under plain SGD, and whether that rank is causally load-bearing rather than incidental.

2.3 Why the readout must be exact, never argmax

Fact 1 holds only under *exact continuous* recovery. Nichani et al. [16] formalize the same linear associative memory $W = \sum_x u_{f(x)} e_x^\top$ and, in a remark to their Theorem 1, give a hand-built *existence* construction: if W is restricted to rank m, random Gaussian mixing lets it store $\approx md$ associations under *discrete argmax* decoding, which tolerates interference. Under that decoding rule a rank-1 matrix can recover $\approx d$ bindings, and Fact 1's necessity collapses. Barnfield et al. [3] sharpen the same argmax-vs-listwise capacity picture with asymptotically exact thresholds for a *static* (non-trained, non-recurrent) memory – useful as an analytic capacity reference, not as a substitute for measuring what a trained model does.

This is the single most consequential design decision in this line of work, and the axis on which we differ from Nichani et al. [16]: we pin the decoder to the literal linear unbind, producing a *continuous* vector,

$$\text{pred} = Z \cdot \text{key}_j,$$

scored by an *absolute* criterion, $\cos(\text{pred}, v_j) > \tau$ – never a K-way argmax among the sample's own values, and never a generic MLP (a generic MLP re-opens the shortcut: non-linearity alone does not force rank use, the central negative-result mechanism in the companion paper). Had we scored by nearest-neighbor over a codebook instead, a rank-1 Z would have "passed" at every K we test, and every result below would be a false negative for the rank hypothesis rather than a genuine measurement of it.

2.4 Closing the position-decomposition escape

A hard bottleneck on the readout is not enough by itself. In any full-attention model, "hold K items" is trivially satisfiable by attending to K raw-input positions at rank-1 each – the *position-decomposition* escape that independently killed an earlier rank story in this program (`rank_aware_v1`: a run at measured effective rank 13.2 was functionally rank-1, composing entirely via position rather than within-matrix spectral structure). We foreclose it architecturally: all information from the K bindings funnels through a single $d \times d$ state Z (P=1), and the decoder is a pure function of $(Z, \text{query key})$ – it never attends to the raw BIND tokens. This is enforced by an explicit attention mask, not assumed, and verified at runtime by a gradient-based *blank-out test*: after the encoder writes Z, the raw-input key/value cache is corrupted and the decode path is confirmed bit-for-bit unchanged across the full decode sequence. A shape check on the mask is not sufficient evidence that a bottleneck holds – only an input-corruption test that the output is provably insensitive to the corrupted path is.

K	1	2	3	4	6	8	10	12	14	16
d=16	2.42	3.01	3.92	4.74	6.40	**8.20**	9.89	11.78	13.47	15.09
d=8	1.65	3.44	–	4.57	–	**7.83**	–	–	–	–

Table 1: Learned effective rank of the unconstrained Z vs. binding count K, mean over ≥ 5 seeds. At d=16, effective rank tracks K almost exactly across the full tested range (Spearman $\rho = 1.0$; the only departure from the pre-registered $[0.7K, 1.3K]$ band is at K=1, 2, where rank modestly exceeds K). At d=8, rank tracks K up to the ceiling (K=8 $\to$ 7.83 $\approx d$). For $K > d$ (not shown; d=16: K=20 $\to$ 11.1, K=24 $\to$ 11.6, K=32 $\to$ 8.6) rank saturates and declines – the expected lossy over-complete HRR regime.

2.5 Architecture and controls

A `BindingEncoder` (a `norm_first` Transformer encoder over the K binding tokens, followed by d learned row-reader latents that cross-attend to produce the d rows of Z) maps the bindings to a single $Z \in \mathbb{R}^{d\times d}$ – permutation-invariant in the bindings, able to express any rank up to d, and not hard-coding the closed-form solution $Z = \sum_j v_j k_j^\top$, so "discovers rank $\approx K$" is a genuine empirical question rather than an architectural given. Default width $h = 64$, 3 layers, 1 refinement iteration; ≈171K parameters at $d = 16$. Loss is $1 - \cos(\text{pred}, \text{target})$. Effective rank (the primary metric, entropy of the normalized singular-value spectrum) is computed via $\text{eigh}(ZZ^\top)$ rather than full SVD for a numerically stable backward pass at degenerate spectra.

The pre-registered controls (full list in the design documents, §8) include: (C1) train-time `force_rank_k` – SVD/eigh-projecting Z to rank k *every training step*, the primary causal test, not post-hoc truncation, which was uninformative in the companion paper's CODI setting; (C3) $\geq$ 5 seeds, with effective rank as the primary metric and stable rank secondary, pre-registered to avoid post-hoc metric shopping; (C4) chance-adjusted per-item recovery; and (C5) held-out generalization on never-seen keys/values. All hypotheses and decision criteria (CONFIRM/FALSIFY bands, e.g. Spearman $\rho(K, \text{eff-rank}) \geq 0.8$ and eff-rank $\in [0.7K, 1.3K]$ for M1) were pre-registered before training.

3 Task D: Rank Tracks K, and Is Causally Necessary

3.1 M1 – learned effective rank tracks K

Table 1 is a 991-run mid-sweep snapshot; a subsequent 1,234-run mega-replication is directionally identical and refines several cells (e.g. d=16: K=8 $\to$ 8.198, K=16 $\to$ 15.083, K=24 $\to$ 10.40, K=32 $\to$ 9.46) without changing the qualitative story, so we report the originally pre-registered table and note the replication for the record.

fr $k \rightarrow$	1	2	3	4	5	6	7	8	16
d=8, K=4	0.0	0.0	0.0	**0.97**	0.99	0.96	–	0.98	–
d=16, K=8	0.0	0.0	–	–	–	0.34	**0.91**	0.76	0.99
d=16, K=12	0.0	0.0	–	–	–	–	–	–	0.9994

Table 2: recovered_frac@0.9 under train-time force-rank k. d=8, K=4 is razor-sharp: force-rank $\leq$ 3 gives exactly 0.0 recovery, force-rank 4 jumps to 0.97. d=16, K=8 and K=12 (intermediate k omitted for space; full grid in the repository) show the same step near their respective K, with mild post-knee non-monotonicity attributable to seed noise and the ≈ 0.936 precision ceiling. A 1,234-run mega-replication refines the K=12 transition zone to a ragged ramp rather than a single clean step (intermediate points fr=11→0.75, fr=13→0.79, fr=14→0.85) – the qualitative knee near $k \approx K$ holds, but the transition is noisier than a two-point table implies.

3.2 M3 – the causal force-rank step lands at $k \approx K$

Effective rank tracking K shows the model *can* build rank-K capacity; it does not show that capacity is used. The primary causal test constrains Z to rank k throughout training (not post-hoc truncation) and measures recovered_frac@0.9 – the fraction of the K bindings recovered above cosine threshold 0.9 (re-registered down from the originally pre-registered τ=0.99 once a calibration run showed the reliably-trainable model plateaus at unconstrained cosine ≈ 0.936, before any force-rank sweep ran).

The signature in Table 2 is the paper's central causal result: constraining rank below K destroys recovery, with a step at $k \approx K$. Together with Table 1, M1 and M3 confirm the pre-registered hypothesis at $d \in \{8, 16\}$: SGD develops rank $\approx K$ *and* that rank is causally necessary. This directly resolves the question the companion paper left open – the earlier rank-blindness was task-specific (ProsQA was rank-1-solvable), not a property of the gradient.

One cell needed a second look. At d=16, K=4, the original 8,000-step budget produced recovered_frac@0.9 = 0.045 even at *full* rank (force-rank 16, i.e. no constraint at all), with the entire force-rank sweep confined to 0.0008–0.134 – initially read as a genuine, seed-count-independent convergence ceiling specific to this cell. An opportunistic probe at 2.5× budget (20,000 steps, 2 seeds, unconstrained) instead reached 0.348/0.483 – a 3–10× jump off the earlier ceiling, with effective rank showing an overshoot-then-compress trajectory (1.3 → 7.8 at step 4K, settling to 4.4–5.0 by 20K). This was not a seed-sparsity artifact; it was a step-budget artifact, and it directly foreshadows the pattern §5 names and quantifies across the rest of this program. Whether 0.35–0.48 is itself further budget-limited or a new, lower-than-K=8/12 ceiling for this specific cell is not resolved by this probe alone.

3.3 The trainability frontier at $d \geq 32$ – revised in §5

At the same 8,000–10,000-step budget used above, $d \geq 32$ appeared to fail outright: effective rank collapsed toward ≈ 1 for every K, and recovered_frac@0.9 was 0.0 for every force-rank – the unconstrained model never learned the write, so there was nothing for truncation to degrade. We initially read this as an optimization/architecture limitation (the same fixed h=64 encoder writing into a Z with 16× more entries at d=128 than d=16) and flagged it as the honest limitation motivating the next experiment. A dedicated follow-up (Stage 0, §5) substantially revises this reading: most of what looked like a trainability wall at $d \geq 32$ is a step-budget artifact, and the frontier that survives extended budget is about converged *exactness*, not whether training moves at all.

4 Task E: Exact Composition and the Invariant-Subspace Mechanism

Task D is a single lookup: no composition. The natural next question is whether the causally-necessary rank-K matrix *composes* correctly – whether Z implements a genuine operator, not just a rank-sufficient lookup table. *Task E* extends the grammar to a permutation-cycle variant: K entities related by a single Hamiltonian K-cycle π, so that querying entity i after h relational hops means recovering $v_{\pi^h(i)}$ via h-fold self-application of the trained operator, $Z^h \cdot k_i$ – no learned per-hop parameters. Training sees hops $H_{\mathrm{train}} = \{1, 2, 3\}$; evaluation adds held-out $H_{\mathrm{test}} = \{4, 5, 6\}$ and depth probes $H_{\mathrm{extra}} = \{7, 21\}$, at $d = 16$ with orthonormal keys.[1]

4.1 Exact composition to depth 21

At K=8, unconstrained rank, 4 of 5 seeds reach recovered_frac@0.9 = 1.00 at *every* tested hop – in-distribution (h=1, 2, 3), held-out (h=4, 5, 6), and both extra probes (h=7, 21) – with zero non-finite training steps. The fifth seed is a partial/still-transitioning outlier whose recovery decays with hop distance (1.00 → 0.16 from h=1 to h=7), the same signature an earlier, budget-starved round showed for its one convergent seed – consistent with a seed still mid-transition, not a new failure mode.

4.2 The depth-amplification mechanism

A force-rank probe at K=8 separates two regimes cleanly. Force-rank $k \in \{8, 9\}$ ($\geq K$, provably sufficient) is *dead*

[1] A general random permutation decomposes into short disjoint cycles, so π^h is periodic in the cycle length and "held-out" hops silently collapse into in-distribution or trivial queries via h mod (cycle length) – a design-gauntlet finding (up to 100% collapse measured at small K) that forced the single-Hamiltonian-cycle construction used here; see §8.

h	predicted $\cos(h)$	measured $\overline{\cos}$	measured rec@0.9
1	0.9317	0.9303	0.996
3	0.9258	0.9259	0.986
5	0.9181	0.9212	0.947
7	0.9089	0.9163	0.881
21	0.7536	0.8206	**0.060**

Table 3: Force-rank k=7 ($K-1$) at K=8: the depth-decay curve, predicted purely from the trained operator's own eigenspectrum with no access to the raw key vectors (a synthetic-key construction exploiting that the true keys decompose into the ideal cycle's eigenbasis with equal $1/\sqrt{K}$ magnitude on every mode), against the measured curve. Full 7-hop table in the repository; every point through h=7 matches to within 0.008. The intuition: dropping one of K=8 eigenmodes leaves per-item cosine near $\sqrt{1-1/8} \approx 0.94$ at shallow depth – comfortably above the 0.9 bar – but 21-fold self-application amplifies the deficiency past threshold.

seed	$\text{eff-rank}(A)$	resid.	$\|C\|$	$\text{eff-rank}(D)$	$\rho(D)$
s1 (exact)	8.000	0.7%	0.024	8.000	1.38
s2 (exact)	7.999	1.9%	0.053	8.000	1.27
s3 (exact)	7.999	1.5%	0.099	8.000	2.86
s4 (exact)	7.999	2.4%	0.059	7.999	1.02

Table 4: Entity-subspace decomposition, K=8 converged seeds (means over 4 eval examples). $\text{eff-rank}(A)$ (max 8) is exact K-recruitment on the task-relevant subspace. Residual is the scale-corrected $\|A - c^*\Pi\|_F/\|c^*\Pi\|_F$ against the ideal cycle Π, after fitting the single isotropic scale c^* that cosine similarity cannot see (raw, scale-naive residuals are 5–180× larger and misleading). $\|C\|$ (leakage into the complement) is $< 2.5\%$ of $\|Z\|$ in every seed. $\text{eff-rank}(D)$ (max 8) shows the complement is itself full-rank, and $\rho(D) \geq 1$ shows it is non-contractive – yet it never corrupts recovery, because no real query trajectory (which always starts in $\mathcal{E}$) ever visits it.

in all 6 tested seeds – stable rank collapses to $\approx$ 1.0–1.15, recovery is 0.00 at every hop, accompanied by numerical instability in the constrained eigh-backward pass, a distinct failure mode from Task D. Force-rank k=7 ($K-1$, provably insufficient) converges in one of three tested seeds, with a revealing signature: in-distribution recovery 1.00/0.99/0.99 (h=1–3), held-out 0.97/0.95/0.92/0.88 (h=4–7), and h=21 = 0.06.

An entity-subspace eigenanalysis (§4.3) confirms this is not a heuristic but an exact spectral fact: *exactly one* eigenmode of the trained operator has the maximum possible phase residual (genuinely absent, not merely mis-rotated) while the other 7 match the ideal cycle to residual 0.001–0.039. Contrast this with the rank-sufficient result of §4.1: seeds with restricted rank provably at K hold cosine 1.00 at h=21 with *zero* decay. Same task, same K, same hop, two rank regimes – one exact through depth 21, one exposed exactly there. **Iteration depth is a sharper spectral-exactness probe than any single-hop cosine tolerance**, and the recoverable-fraction metric is far more sensitive to depth than the mean cosine (Table 3: $\overline{\cos}$ falls only 0.93 $\rightarrow$ 0.82 from h=7 to h=21 while rec@0.9 collapses 0.88 $\rightarrow$ 0.06), because the per-item phase-alignment distribution widens under repeated composition rather than merely shifting.

4.3 The invariant-subspace decomposition

Whole-matrix effective rank of the converged K=8 solutions is $\approx$ 14.7–15.6 out of d=16 – essentially full ambient dimension, not $\approx K$. Taken at face value this contradicts Task D's M1 pattern. It does not: the whole-matrix number is the wrong instrument.

Recovering the K-dimensional entity subspace $\mathcal{E}$ from the analytic ideal cycle's own SVD (top-K left singular vectors), we block-decompose the trained Z in the $[\mathcal{E}, \mathcal{E}^\perp]$ basis: $A = U^\top Z U$ (the restricted operator), $B = U^\top Z V$ and $C = V^\top Z U$ (cross-subspace leakage), and $D = V^\top Z V$ (the complement). By induction, a query starting in $\mathcal{E}$ stays entirely within $\mathcal{E}$ under repeated self-application – $U^\top Z^h u = A^h a$ exactly, independent of B and D – *iff* $C \approx 0$.

Table 4 answers the M1-transfer question cleanly: **restricted to the entity subspace, "rank tracks K" holds exactly** – it was the whole-matrix measurement that was uninformative, not SGD's rank discipline. D's effective rank ($\approx K$ out of its own $d-K$ dimensions) is exactly the missing piece of the whole-matrix inflation (K in A plus $d-K$ in $D \approx d$). D is neither task-structured nor i.i.d. noise – its singular-value spectrum is nearly flat (condition number 1.01–1.75, an order of magnitude flatter than a random Gaussian matrix of the same norm) – consistent with it being whatever the encoder's untrained-in-this-direction output defaults to, rather than anything the loss shaped. Dead force-rank seeds fail this test completely: $\text{eff-rank}(A) \approx 1.00$ even restricted to the entity subspace, confirming the collapse is genuine, not an artifact of measuring the wrong matrix.

4.4 The K-wall was also a budget artifact, with one genuine boundary case

At the same 40,000-step budget, $K \in \{12, 16\}$ (unconstrained rank) appeared dead – 0/5 and 0/5 seeds converged. Doubling the budget to 80,000 steps reversed this: K=12 is 3/3 exact (recovery 1.00 at every hop through h=21 in 2 of 3 seeds, 0.9999 in the third), and K=16 is 2/3 converged (near-exact through h=7, with the depth-amplification signature of §4.2 appearing at h=21 in an *unconstrained*-rank seed for the first time, not only in the deliberately rank-starved case). Onset windows are seed-dependent and grow with K: ≈12–40K steps at K=8; ≈45–75K at K=12/16.

The subspace decomposition replicates the K=8 story at K=12 (restricted rank 99.97–100.0% of K, leakage $< 2.1\%$ of $\|Z\|$) and at K=16's two converged seeds (99.76–99.9%), with one structural caveat: at K=16=d, the entity subspace *is* the full ambient space, so B, C, D are 0×0 by construction for any Z, trained or not – "leakage ≈ 0" at K=d is a tautology of the geometry, not a finding about what SGD learned, and should not be cited alongside the genuine low-leakage results at $K < d$.

K=16's remaining stuck seed is the one place a further budget extension did *not* help: 3 seeds run to 120,000 steps (1.5× the 80K round) are all dead – mean_cos at noise level (-0.003 to 0.066) at every hop, effective rank 1.5–3.0, no transition signature in the training loss at any point in the extended budget. Combined across both budgets, K=16's tally is $2/5$ converged, $3/5$ dead – the one cell in this document where the entity subspace is the entire ambient space, with no orthogonal complement providing slack for a marginal write to grow into before an exact solution is required. We flag this explicitly as a boundary-case scaling observation, distinct from §5's three-budget-artifacts pattern: unlike every other instance of that pattern, this specific extension rescued zero additional seeds.

5 The Exactness Frontier, and When to Trust a "Dead" Cell

§3.3 left $d \geq 32$ looking like a trainability wall. *Stage 0* is a dedicated diagnostic campaign, pre-registered with a formal pass bar (recovered_frac@0.9 > 0.7 at d=32) and a weaker diagnostic-pass condition (a measured, falsifiable verdict on every named candidate mechanism, independent of whether the formal bar is met).

5.1 The three-budget-artifacts pattern

At 2.5× Task D's original 8,000-step budget (20,000 steps), all 6 baseline d=32 seeds ($K \in \{8, 16\}$, 3 seeds each) transition: effective rank is flat near 1.0 for 6,000–10,000 steps, then climbs sharply, in every seed, somewhere in a 6–16K window – directly falsifying the reading that the original 8K-step snapshot was a converged state. A screen of four candidate interventions (LR warmup, orthogonal initialization, μP width scaling, self-attention among row-queries) at matched 16K-step budget found *zero* that beat the unconstrained baseline; step budget alone, extended to 40,000 steps, then reached recovered_frac@0.9 $= 0.445$ in the best seed – higher than any screened intervention achieved at any checkpoint.

This is the third instance of the same signature in this program (loss flat near the trivial-solution value for most of a budget, then a sharp, seed-stochastic collapse, misread as a wall at the original budget):

1. Task E, K=8: $1/5$ seeds converged at 20K steps; $4/5$ at 40K (§4.1).
2. Stage 0, $d \geq 32$: the original 8K-step wall (rank ≈ 1, recovery ≈ 0) reversed to 17/17 baseline seeds transitioning reliably across Wave 0, an extended-budget arm, and a 100K-step probe, onset 6–16K steps (this section).

d	K=$d/4$	n seeds	budget	$\overline{\cos}$
16	4	Task D/E (§3–4)	40–80K	≈ 1.00 (exact)
32	8	3	100K	0.877–0.915
48	12	1	100K	0.7196
64	16	1	150K	0.3882

Table 5: The K=$d/4$ exactness frontier – the cleanest slice: monotone, no overlap between adjacent d, and every cell's trajectory confirmed flat (oscillating within a fixed window over the trajectory's final quarter, not still climbing) at the budget shown before being read as converged.

3. Task E, $K \in \{12, 16\}$: $0/5$ and $0/5$ "dead" at 40K steps; $3/3$ and $2/3$ converged at 80K, onset 45–75K (§4.4).

Rule: a cell should not be called dead without retesting at 2–2.5× its original budget. This is a program-level methodological finding, not a per-experiment footnote, and we state the counter-caution with equal weight: **the rule is necessary, not sufficient.** Stage 0's own 100K-step probe (10× the original budget) at d=32 produced trajectories that are visibly *flat* in their final 60,000+ steps – recovered_frac@0.9 oscillating in fixed windows (e.g. $[0.596, 0.632]$ for one K=8 seed across the final six 2,000-step checkpoints) rather than climbing – and the best result across the entire campaign, 0.6639, still falls short of the pre-registered > 0.7 bar. This is a genuine, budget-independent plateau: the formal CONFIRM criterion *fails* at d=32 even though the diagnostic-pass condition succeeds (every named mechanism – architectural cap, row-query collision, dead initialization, undertraining, late transition – received a measured, falsifiable verdict rather than a shrug). Re-testing at extended budget is mandatory before declaring a cell dead; it is not a guarantee the cell will turn out alive.

5.2 Exactness degrades monotonically with d

Table 5 is the clean version of the frontier: fixing K=$d/4$ and scaling d at fixed encoder width h=64, converged cosine recovery falls monotonically from essentially exact at d=16 to 0.39 at d=64, with every point confirmed converged (not truncated mid-transition) via flat trailing checkpoints. A parallel K=$d/2$ slice is directionally consistent but noisier and confounded (budget is not held fixed across d; the single d=48, K=24 seed nearest completion is a still-climbing floor, not yet a plateau, so it should not be read as a frontier reversal). Recovered-fraction, the exact-match metric, is qualitatively harsher than cosine: it is 0.0 at every d=64 seed tested, against nonzero cosine everywhere.

A second gradation compounds this one. At d=32, effective rank still tracks K closely (91–97% of target) even though exact recovery plateaus short of the bar – rank recruitment and final-readout exactness are separable there. At $d \geq 48$, rank recruitment itself becomes partial and the fraction shrinks with K (e.g. at d=64: K=16 $\to$ 21% of target,

K=32 $\to$ 32–57%, with wider seed spread than d=32's own K=16 baseline) – a distinct phenomenon from the d=32 read, worth tracking as future scaling-law data rather than a characterized ceiling.

This is not the same as the one proven architectural cap. A separate, purely algebraic fact – $\mathrm{rank}(Z) \leq h+1$ for the h-width row-reader encoder used throughout – gives a hard ceiling of 65 at h=64. This binds only at d=128, $K > 65$ and is untouched by anything in this section; every $d \in \{32, 48, 64\}$ cell reported above sits well under its own architectural cap, so the empirical exactness frontier is a genuinely separate phenomenon from this proven bound, not a restatement of it.

One unresolved non-monotonicity. At d=32, exact-recovery performance is *not* monotone in K at fixed budget: K=4 plateaus at recovered_frac@0.9 $\approx$ 0.148 (100K steps, flat tail) despite healthy cosine (0.830, comparable to K=8's 0.877–0.914), while K=8 reaches 0.41–0.65 and K=16 falls back to 0.248 – K=8 = $d/4$ outperforms both neighbors in the tested grid. This survives a 5$\times$ and then 10$\times$ budget extension past the original 8K sweep, so it is a converged phenomenon, not an undertraining artifact, and it remains open and unexplained.

6 Related Work

Factual recall via associative memory. Nichani et al. [16] is the paper this work is in most danger of being read as a rank-flavored variant of, and §2.3 addresses it head-on: their rank-m capacity result is a hand-built existence construction under interference-tolerant argmax decoding, not a measurement of the rank gradient descent discovers, a necessity bound for exact recovery, or a causal ablation; their transformer has full self-attention over raw tokens, with no hard single-state bottleneck. Barnfield et al. [3] (with two shared authors) sharpen the argmax-vs-listwise capacity threshold for a *static* rank-constrained memory with asymptotically exact constants – a useful analytic reference point, not an object that is trained, composed, or causally ablated.

Rank of trained fast-weight states. Nazari and Rusch [15] and Sun et al. [23] measure the same effective-rank metric on the linear-attention state $S_t = \sum_t v_t k_t^\top$ and prove the *upper* bound $\mathrm{rank}(S_t) \leq t$, in pretrained LLMs with uncontrolled write count on real text; neither has a necessity theorem or a training-time rank-k ablation. This work is the mirror image: controlled K, a *lower* bound, and a causal force-rank curve, in a from-scratch synthetic setting with provable ground truth.

Eigenstructure and state-tracking. Composability under repeated application is gated by eigenvalue sign and phase, not rank magnitude alone [7, 21]. This motivates the eigenstructure-fidelity view used throughout §4: a rank-K matrix with the wrong eigenstructure can pass one-hop recall while failing composition, exactly the failure mode §4.2 isolates for force-rank $K-1$. We are careful not to conflate two different objects named "rank" in this lineage: DeltaProduct's rank is a property of the per-token *transition matrix* $A_t = I - \beta_t k_t k_t^\top$ applied to the state, whereas the quantity measured throughout this paper is the *state's own* rank, $\mathrm{rank}(Z)$ or $\mathrm{rank}(S_t)$ – the capacity of what has been written so far. Conflating these is a natural point of reviewer confusion.

Matrix-valued recurrent state and compositional transfer – the closest prior art. Mishra et al. [14] report the closest existing result to §4: a matrix-valued recurrent state (M^2RNN) achieving near-perfect state-tracking generalization on S_3-permutation composition, trained at sequence length 128 and evaluated at 512 – the same compose-beyond-the-training-horizon shape as this paper's Task E. Our delta, led with what their own results show: (i) their own comparison table has a plain vector-state GRU matching the matrix model on the same task, so their result does not by itself show matrix rank, or matrix structure at all, is load-bearing; (ii) their state-tracking theorem is representational (an existence/expressivity result), not a rank-necessity bound, and nothing in their setup measures what rank SGD's solution actually uses; (iii) they run no causal rank intervention – no train-time rank forcing, no truncation ablation. §4 is the mirror image on all three: a task where $\mathrm{rank}(Z) \geq K$ is provable under the pinned readout (§2.2), a train-time causal force-rank arm (§4.2), and a subspace-restricted verification that the rank actually recruited is exactly K (§4.3).

Theoretical capacity of continuous CoT. Zhu et al. [25] and Gozeten et al. [6] give existence constructions for parallel reasoning in continuous latents, bounded by embedding dimension; both are capacity results with small empirical demonstrations, not measurements of what a trained objective actually recruits. Rizvi-Martel et al. [18] push back empirically, showing a fine-tuned COCONUT model reaches high ProsQA accuracy without latent tokens at all – the direct empirical motivation for our companion paper [10] and, transitively, for this one: this paper does not adjudicate whether *real* reasoning models superpose, but it does establish that a matrix latent *can* be driven to functional, causally necessary rank K under gradient training, on a task chosen so that "functional" is provable rather than inferred from accuracy alone.

Standing caution on compositional generalization. Dziri et al. [4] and Wang et al. [24] are counter-evidence and caution for any compositional-generalization claim: transformers can grok in-distribution composition yet fail on novel combinations, or reduce to linearized subgraph matching that looks like genuine multi-step reasoning but is not. Our claim in §4 is deliberately narrower than a general compositionality claim – it is about the specific rank-K matrix state Task D showed is causally necessary, verified by an exact eigenstructure decomposition against a known analytic target, not an inference from held-out accuracy alone.

Binding-as-outer-product and VSA capacity. Z is literally a Tensor Product Representation [11, 19, 22]. The Vector Symbolic Architecture tradition [5, 17] studies a different capacity notion – SNR and bundling/codebook factorization, not matrix rank – which is why the exact-rank framing here is a distinct, cleaner question, and why the $K > d$ regime (not gated on in this paper) is expected to be lossy in the HRR sense rather than a rank-necessity failure. Arora et al. [2]'s MQAR is the standard synthetic associative-recall family this paper's grammar resembles; it varies state dimension against a fixed pair count and reports end-task accuracy, not rank spectra. Merrill et al. [12] bounds SSM state expressivity via communication complexity – a size/complexity question, architecture- and decoding-agnostic, that cannot by construction isolate rank as the causal quantity the way a fixed-d^2, varying-rank ablation can.

7 Discussion

Scope: synthetic tasks, ≤1M parameters. Every experiment in this paper is small by contemporary standards: the largest model tested has ≈ 183K parameters (d=64), and every task is a synthetic associative-memory or composition grammar with no natural-language content. This project's own working practice treats results at this scale with caution for good reason – at a few hundred thousand parameters on natural text, models barely learn unigram statistics, and no claim about reasoning, generalization, or abstract thought is licensed by such a model's behavior alone. That caution does not straightforwardly transfer here, and it is worth being explicit about why. A behavioral claim on real text ("this model reasons") has no ground truth to check against except more behavior, so a small model's poor performance is genuinely uninformative about the mechanism in a larger model. The claims in §3–4 are different in kind: $\mathrm{rank}(Z) \geq K$ is a proven necessity for the exact solution at *any* scale, so finding effective rank $\approx K$, a causal step at $k \approx K$, and an entity-subspace-restricted operator that is exactly the analytic target are not behavioral proxies for a mechanism – they are direct measurements of a mechanism against a known, provable ground truth. What small scale does still cost us is generality: we do not know whether the same mechanism is recruited at 10M+ parameters, on real data, or under a training objective with no comparably clean ground truth to check against. That is precisely the transfer question §5's exactness frontier and this project's own sequencing treat as the next, separate step, gated on the results reported here – not yet run.

What this does not show. Task D and Task E are deliberately abstract, chosen for provability rather than realism. They do not show that matrix-valued rank helps performance on a real reasoning benchmark, that the mechanism survives standard-scale pretraining, or that it competes with any production architecture on accuracy or throughput. §6's discussion of Dziri et al. [4] and Wang et al. [24] applies directly here: a mechanism shown to compose exactly on a synthetic cycle with known structure is not evidence that the same mechanism generalizes compositionally on tasks without that structure.

Concurrent, preliminary: DeltaNet. A separate, still-running campaign asks whether the same causal-rank story holds on a production fast-weight architecture – vanilla, single-Householder-per-token DeltaNet [20], whose delta rule $S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$ only ever writes within $\mathrm{span}(\{k_j\})$ by construction. On the same Task-E-style composition grammar, the unconstrained arm reaches $\mathrm{recovered\_frac@0.9} = 1.00$ at every tested hop (1–7, 21) at all four tested (d, K) cells within $2.5\times$ tier-default step budget, with zero training instability – a materially different outcome from the bespoke encoder used throughout this paper, which plateaus sub-exact at the same $(d, K) = (64, 32)$ operating point after 150K steps. Architecturally, entity-subspace rank equals whole-matrix rank exactly, with no inflation: the delta rule has no orthogonal complement for an optimizer to leave untouched, so §4.3's subspace-restriction fix is unnecessary here by construction rather than by a fortunate optimization outcome. This is consistent with, and architecturally cleaner than, this paper's Task E result. It is *not* yet a completed causal claim: the force-rank staircase that would show a $k{=}K{-}1$ collapse (this paper's central test in §3.2 and §4.2) is mid-campaign on DeltaNet – a $k{=}K$ (provably sufficient) cell is climbing but has not reached ceiling within its tier budget, and the $k{=}K{-}1$ comparison has not run. We report this paragraph only to bound what is already visible in that campaign; the full DeltaNet result, including the causal staircase, is a separate paper.

Falsification. For the core causal claim (§3.2, §4.2): a force-rank-$(K{-}1)$ model reaching $\geq 0.9\times$ ceiling accuracy at multiple (d, K) operating points would falsify it. This has not happened – the one force-rank-$(K{-}1)$ Task E seed that converged at all showed exactly the predicted depth-decay signature (Table 3), not ceiling accuracy. For the exactness-frontier claim (§5): a $d \geq 32$ cell reaching the pre-registered > 0.7 bar at any tested budget would revise it toward "trainability, not exactness, is the limit"; this has not happened through 100K–150K steps with confirmed-flat trailing checkpoints. For the concurrent DeltaNet result: a completed Wave-B force-rank grid showing no accuracy step between $k{=}K{-}1$ and $k \geq K$ even at extended budget would be the relevant negative there, distinct from – and, per §5's pattern, requiring the same budget-extension discipline as – every negative reported in this paper.

A standing caution about our own negative results. Three independent axes of this program each declared a cell dead at its first tested budget before a 2–2.5× extension reversed the verdict (§5.1). A reader evaluating any single "wall" claim in this literature, including ones in our own earlier project documents that this paper's numbers supersede, should ask whether it was re-tested at extended budget before being treated as final.

8 Conclusion

Our companion negative result left one question open: is a matrix latent's rank ignored by the gradient in general, or only when nothing rewards it? Closing every shortcut that could make a model look rank-aware without needing rank – argmax decoding, position-decomposition, and a step budget too short to see a late transition – the answer is that the gradient does see rank, when a task provably forces it. At $d \in \{8, 16\}$, gradient descent trained from scratch recruits effective rank $\approx K$ and makes it causally necessary, with a step at $k \approx K$ sharp enough to be razor-clean at the smallest tested cell. That rank composes: a rank-sufficient trained operator self-applies exactly to depth 21, and an entity-subspace decomposition shows this is not a lucky lookup table but a genuine, restricted-rank-exactly-K realization of the target permutation, with an invisible, unconstrained complement absorbing everything the loss never asked it to control. Depth itself is a sharper rank probe than any single-hop tolerance: a deliberately rank-starved operator's entire depth-decay curve is predictable from its own spectrum, no raw keys required. And what looked like a hard trainability wall at larger d is substantially a step-budget artifact – three independent axes of this program made the same mistake before a 2–2.5× budget extension corrected it – with a genuine, narrower frontier surviving underneath: not whether training moves, but how exactly it converges as d grows. Every number here traces to a specific run, seed, and step count, and every claim above has a stated falsification condition. The transfer question – whether any of this helps reasoning on real data – is not addressed here by design, and is the explicit next step this line of results gates.

Reproducibility

All training, evaluation, and analysis code is released at `https://github.com/saml212/learned-representations` under `matrix-thinking/chapter2/`.

Code, by experiment. Task D: `task_d.py` (grammar generator), `model_v4.py` (encoder + pinned unbind), `rank_utils.py` (eigh-stable effective/stable rank), `run_task_d.py` (train/eval/smoke), `run_overnight.py` (perpetual, resumable 8×H100 orchestrator). Task E: `task_e.py`, `run_task_e.py`, `run_task_e_sweep.py` (reuses Task D's encoder and rank utilities verbatim), `eigen_utils.py` (eigenstructure-fidelity metric, C8), `analyze_zdump.py` (numpy + stdlib only, no GPU required; regenerates every number in §4.3–4.4 from archived Z-dumps). Stage 0: `run_stage0.py`, `run_stage0_sweep.py`.

Pre-registrations. `TASK_D_PREREGISTRATION.md` (hypothesis, proof, controls, decision criteria, logged before any training); `NEXT_EXPERIMENT_DESIGN.md` (Task E design and its H_{extra} residue-collision addendum); `STAGE0_DESIGN.md` (the exactness-frontier diagnostic, formal and diagnostic pass criteria). Every pre-registered decision criterion and every deviation from it (e.g. the τ=0.99 $\rightarrow$ 0.9 metric re-registration in §3.2, logged before the sweep; Stage 0's Wave B skip in favor of the 100K probe, §5.1) is documented inline in these files with a timestamp, not silently applied.

Audit trail. Design and code passed an adversarial gauntlet before any GPU run for each experiment: a task-shortcut attack (which killed an earlier crossover design and produced Task D; which caught the injectivity-tolerance and general-permutation bugs that would have silently broken Task E's held-out-hop claim before any compute ran), a novelty check, three rounds of code audit, and two rounds of runner audit (`gauntlet/`), plus a runtime smoke gate (degenerate-spectrum backward, blank-out leak test, resolution pre-flight).

Hardware and compute. Brev 8×H100 80GB cluster, torch 2.12/cu13. Task D: ≈56 GPU-h for the 991-run snapshot reported in §3, plus a subsequent 1,234-run mega-replication. Task E: the 40K-step calibration round (33 runs), the 80K-step K-wall round (6 runs, 14.5 GPU-h), and the 120K-step K=16 completion wave (3 runs). Stage 0: 27.66 GPU-h core (Wave -1/0/A, an extended-budget arm, and the 100K probe, against a pre-registered 60 GPU-h budget), plus an 11.36 GPU-h post-closure exactness-frontier follow-on (d=64 and d=48 points). Cumulative project grant burn at time of writing: ≈320 of a 1,600 GPU-h grant. The concurrent DeltaNet result in §7 is ≈10 GPU-h so far, Waves -1/0 only, explicitly partial.

Data. Raw per-run JSON results (dense training-step checkpoints, not just final values – every trajectory-based claim in this paper, e.g. "flat, not climbing" vs. "still climbing at cutoff," is verified from the full checkpoint array, not a summary rollup) are archived under `experiment-runs/`, in five dated subdirectories: `2026-07-02_task_e_zdump`, `2026-07-02_task_e_80k_kwall`, `2026-07-02_task_e_120k_k16`, `2026-07-02_stage0_waves`, and `2026-07-02_deltanet_waves`. All numbers in this paper trace to a specific archived JSON, run tag, and seed; none are recomputed or estimated for this write-up.

References

[1] James A. Anderson. A simple neural network generating an interactive memory. *Mathematical Biosciences*, 14(3–4):197–220, 1972.

[2] Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. Zoology: Measuring and improving recall in efficient language models. *arXiv preprint arXiv:2312.04927*, 2023.

[3] Nicholas Barnfield, Juno Kim, Eshaan Nichani, Jason D. Lee, and Yue M. Lu. Sharp capacity thresholds in linear associative memory: From winner-take-all to listwise retrieval. *arXiv preprint arXiv:2605.05189*, 2026.

[4] Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jian, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. Faith and fate: Limits of transformers on compositionality. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2305.18654.

[5] E. Paxon Frady, Denis Kleyko, and Friedrich T. Sommer. Resonator networks, 1: An efficient solution for factoring high-dimensional, distributed representations of data structures. *Neural Computation*, 32(12):2311–2331, 2020. arXiv:1906.11684.

[6] Ata Gozeten, M. Emre Ildiz, Yiding Zhang, Hrayr Harutyunyan, Ankit Singh Rawat, and Samet Oymak. Continuous chain of thought enables parallel exploration and reasoning. *arXiv preprint arXiv:2505.23648*, 2025.

[7] Riccardo Grazzi, Julien Siems, Jörg K. H. Franke, Arber Zela, Frank Hutter, and Massimiliano Pontil. Unlocking state-tracking in linear RNNs through negative eigenvalues. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2411.12537.

[8] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. In *arXiv preprint arXiv:2412.06769*, 2024.

[9] Teuvo Kohonen. Correlation matrix memories. *IEEE Transactions on Computers*, C-21(4):353–359, 1972.

[10] Sam Larson. The gradient does not see rank: Rank-indifference in matrix-CODI on ProsQA. In *ICML 2026 Workshop on Mechanistic Interpretability*, 2026. Companion paper.

[11] R. Thomas McCoy, Tal Linzen, Ewan Dunbar, and Paul Smolensky. RNNs implicitly implement tensor product representations. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1812.08718.

[12] William Merrill, Jackson Petty, and Ashish Sabharwal. The illusion of state in state-space models. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2404.08819.

[13] Francesco Mezzadri. How to generate random matrices from the classical compact groups. In *Notices of the American Mathematical Society*, 2007.

[14] Mayank Mishra, Shawn Tan, Ion Stoica, Joseph Gonzalez, and Tri Dao. M^{2}RNN: Non-linear RNNs with matrix-valued states for scalable language modeling. *arXiv preprint arXiv:2603.14360*, 2026.

[15] Pedram Nazari and T. Konstantin Rusch. The key to state reduction in linear attention: A rank-based perspective. *arXiv preprint arXiv:2602.04852*, 2026.

[16] Eshaan Nichani, Jason D. Lee, and Alberto Bietti. Understanding factual recall in transformers via associative memories. In *International Conference on Learning Representations (ICLR), Spotlight*, 2025. arXiv:2412.06538.

[17] Tony A. Plate. Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6(3):623–641, 1995.

[18] Michael Rizvi-Martel, Marius Mosbach, and Guillaume Rabusseau. The illusion of superposition? a principled analysis of latent thinking in language models. *arXiv preprint arXiv:2604.06374*, 2026.

[19] Imanol Schlag, Paul Smolensky, Roland Fernandez, Nebojsa Jojic, Jürgen Schmidhuber, and Jianfeng Gao. Enhancing the transformer with explicit relational encoding for math problem solving. In *arXiv preprint arXiv:1910.06611*, 2019.

[20] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, 2021. arXiv:2102.11174.

[21] Julien Siems, Timur Carstensen, Arber Zela, Frank Hutter, Massimiliano Pontil, and Riccardo Grazzi. DeltaProduct: Improving state-tracking in linear RNNs via householder products. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2502.10297.

[22] Paul Smolensky. Tensor product variable binding and the representation of symbolic structures in connectionist systems. *Artificial Intelligence*, 46(1–2):159–216, 1990.

[23] Ang Sun, Hao Zhang, Hang Zhou, Yifei Ma, Yujia Qin, Ting Su, Yang Liu, Zhen Ma, Jian Xu, Jianfeng Gao, Jiawei Hao, and Ruoxi He. State rank dynamics in linear attention LLMs. *arXiv preprint arXiv:2602.02195*, 2026.

[24] Boshi Wang, Xiang Yue, Yu Su, and Huan Sun. Grokked transformers are implicit reasoners: A mechanistic journey to the edge of generalization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.15071.

[25] Hanlin Zhu, Shibo Hao, Zhiting Hu, Jiantao Jiao, Stuart Russell, and Yuandong Tian. Reasoning by superposition: A theoretical perspective on chain of continuous thought. *arXiv preprint arXiv:2505.12514*, 2025.